

# GASTROINTESTINAL HORMONES

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## 1. GASTRIN

- **Structure** : Peptide with **34 amino acid (AA) residues**.
- **Site of Secretion** :—
  - Secreted mainly by **G-cells** of pyloric glands of stomach.
  - Also secreted by **TG-cells** in stomach, duodenum & jejunum.
  - In fetus: Islets of Langerhans also secrete gastrin.
  - Secreted from stomach (*during gastric phase*) and from small intestine (*during intestinal phase*) of gastric secretion.

## STIMULANT FOR SECRETION

- a. Presence of food in stomach.
- b. Stimulation of local nervous plexus in stomach & small intestine (SI).
- c. **Vagovagal reflex** during gastric phase of gastric secretion:

Gastrin-Releasing Polypeptide (GRP) released at vagal nerve ending



Causes secretion of Gastrin by stimulating G-cells

## ACTIONS OF GASTRIN

- a. **Stimulates gastric glands** : Secretes gastric juice rich in HCl + pepsin.
- b. **Accelerates gastric motility**.
- c. **Promotes growth of gastric mucosa**.
- d. **Stimulates secretion of pancreatic juice** (*rich in enzymes*).
- e. **Stimulates islets of Langerhans** in pancreas to release pancreatic hormones.

## 2. SECRETIN

- **Discovery** : 1<sup>st</sup> ever hormone discovered (*by Starling & Bayliss, 1902*).
- **Structure** : Peptide hormone with **27 AA residues**.
- **Site of Secretion** : Secreted by **S-cells** of duodenum, jejunum & ileum.
- **Activation** :



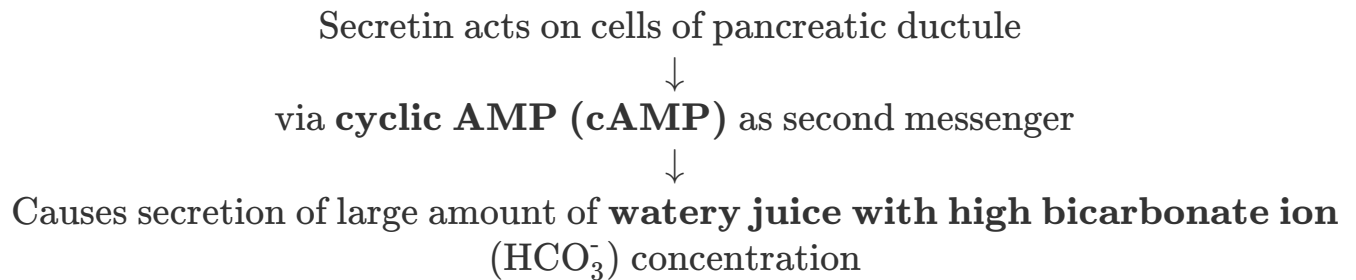
## STIMULANT FOR SECRETION

- a. Release & activation of prosecretin by **acid chyme** entering duodenum from stomach.
- b. Products of protein digestion.

# ACTIONS OF SECRETIN

## Major Actions:

- Stimulates **exocrine pancreatic secretion**:



## Other Actions:

- a. **Inhibits secretion of gastric juice.**
- b. **Inhibits motility of stomach.**
- c. **Causes constriction of pyloric sphincter.**
- d. **Increases potency** of the action of cholecystokinin on pancreatic secretion.

## 3. CHOLECYSTOKININ (CCK / CCK-PZ)

- **Nomenclature** : Also called **Cholecystokinin–Pancreozymin (CCK-PZ)**.
- **Structure** : Made up of **39 AA residues**.
- **Site of Secretion** : Secreted by **I-cells** in mucosa of duodenum & jejunum, and small quantities in ileum.

## STIMULANT FOR SECRETION

- Presence of chyme containing digestive products of **fats & proteins** (*namely Fatty Acids [FA], Peptides, and Amino Acids [AA]*) in upper part of small intestine.

## ACTIONS OF CHOLECYSTOKININ

- a. **Contracts gallbladder** to expel stored bile.

- b. **Stimulates exocrine pancreatic secretion** (*enzyme-rich juice*).
- c. **Increases secretion of enterokinase.**
- d. **Inhibits gastric motility.**
- e. **Increases motility of intestine.**
- f. **Induces drug tolerance to opioids.**
- g. **Accelerates the activity of secretin** to produce alkaline pancreatic juice.

# GASTROINTESTINAL HORMONES

## SUMMARY

| Hormone                      | Source of Secretion                                                                                                                                                                       | Actions                                                                                                                                                                                                                                                                                                    |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gastrin</b>               | <ul style="list-style-type: none"> <li>• G cells in stomach</li> <li>• TG cells in GI tract</li> <li>• Islets in fetal pancreas</li> <li>• Anterior pituitary</li> <li>• Brain</li> </ul> | <ul style="list-style-type: none"> <li>• Stimulates gastric secretion and motility</li> <li>• Promotes growth of gastric mucosa</li> <li>• Stimulates release of pancreatic hormones</li> <li>• Stimulates secretion of pancreatic juice</li> <li>• Stimulates secretion of pancreatic hormones</li> </ul> |
| <b>Secretin</b>              | <ul style="list-style-type: none"> <li>• S cells of small intestine</li> </ul>                                                                                                            | <ul style="list-style-type: none"> <li>• Stimulates secretion of watery and alkaline pancreatic secretion</li> <li>• Inhibits gastric secretion and motility</li> <li>• Constricts pyloric sphincter</li> <li>• Increases potency of cholecystokinin action</li> </ul>                                     |
| <b>Cholecystokinin (CCK)</b> | <ul style="list-style-type: none"> <li>• I cells of small intestine</li> </ul>                                                                                                            | <ul style="list-style-type: none"> <li>• Contracts gallbladder</li> <li>• Stimulates pancreatic secretion with enzymes</li> <li>• Accelerates secretin activity</li> <li>• Increases enterokinase secretion</li> <li>• Inhibits gastric motility</li> </ul>                                                |

| Hormone                                        | Source of Secretion                                                                                                                               | Actions                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                |                                                                                                                                                   | <ul style="list-style-type: none"> <li>Increases intestinal motility</li> <li>Augments contraction of pyloric sphincter</li> <li>Suppresses hunger</li> <li>Induces drug tolerance to opioids</li> </ul>                                                                                                                                        |
| <b>Gastric Inhibitory Peptide (GIP)</b>        | <ul style="list-style-type: none"> <li>K cells in duodenum and jejunum</li> <li>Antrum of stomach</li> </ul>                                      | <ul style="list-style-type: none"> <li>Stimulates insulin secretion</li> <li>Inhibits gastric secretion and motility</li> </ul>                                                                                                                                                                                                                 |
| <b>Vasoactive Intestinal Polypeptide (VIP)</b> | <ul style="list-style-type: none"> <li>Stomach</li> <li>Small and large intestines</li> </ul>                                                     | <ul style="list-style-type: none"> <li>Dilates splanchnic (peripheral) blood vessels</li> <li>Inhibits HCl secretion in gastric juice</li> <li>Stimulates secretion of succus entericus</li> <li>Relaxes smooth muscles of intestine</li> <li>Augments acetylcholine action on salivary glands</li> <li>Stimulates insulin secretion</li> </ul> |
| <b>Glucagon</b>                                | <ul style="list-style-type: none"> <li><math>\alpha</math>-cells in pancreas</li> <li>A cells in stomach</li> <li>L cells in intestine</li> </ul> | <ul style="list-style-type: none"> <li>Increases blood sugar level</li> </ul>                                                                                                                                                                                                                                                                   |
| <b>Glicentin</b>                               | <ul style="list-style-type: none"> <li>L cells in duodenum and jejunum</li> </ul>                                                                 | <ul style="list-style-type: none"> <li>Increases blood sugar level</li> </ul>                                                                                                                                                                                                                                                                   |
| <b>Glucagon-like Polypeptide-1 (GLP-1)</b>     | <ul style="list-style-type: none"> <li><math>\alpha</math>-cells in pancreas</li> <li>Brain</li> </ul>                                            | <ul style="list-style-type: none"> <li>Stimulates insulin secretion</li> <li>Inhibits gastric motility</li> </ul>                                                                                                                                                                                                                               |
| <b>GLP-2</b>                                   | <ul style="list-style-type: none"> <li>L cells in ileum and colon</li> </ul>                                                                      | <ul style="list-style-type: none"> <li>Suppresses appetite</li> </ul>                                                                                                                                                                                                                                                                           |
| <b>Somatostatin</b>                            | <ul style="list-style-type: none"> <li>Hypothalamus</li> <li>D cells in pancreas</li> <li>D cells in stomach and small intestine</li> </ul>       | <ul style="list-style-type: none"> <li>Inhibits secretion of growth hormone</li> <li>Inhibits gastric secretion and motility</li> <li>Inhibits secretion of pancreatic</li> </ul>                                                                                                                                                               |

| Hormone                            | Source of Secretion                                                                                                                            | Actions                                                                                                                                                                                                                                               |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    |                                                                                                                                                | juice<br>• Inhibits secretion of GI hormones                                                                                                                                                                                                          |
| <b>Pancreatic Polypeptide (PP)</b> | <ul style="list-style-type: none"> <li>• PP cells in pancreas</li> <li>• Small intestine</li> </ul>                                            | <ul style="list-style-type: none"> <li>• Increases secretion of glucagons</li> <li>• Decreases pancreatic secretion</li> </ul>                                                                                                                        |
| <b>Peptide YY</b>                  | <ul style="list-style-type: none"> <li>• L cells of ileum and colon</li> </ul>                                                                 | <ul style="list-style-type: none"> <li>• Inhibits gastric secretion and motility</li> <li>• Reduces secretion of pancreatic juice</li> <li>• Inhibits intestinal motility and bowel passage</li> <li>• Suppresses appetite and food intake</li> </ul> |
| <b>Neuropeptide Y</b>              | <ul style="list-style-type: none"> <li>• Ileum and colon</li> <li>• Brain and autonomic nervous system (ANS)</li> </ul>                        | <ul style="list-style-type: none"> <li>• Increases blood flow in enteric blood vessels</li> </ul>                                                                                                                                                     |
| <b>Motilin</b>                     | <ul style="list-style-type: none"> <li>• Mo cells in stomach and intestine</li> <li>• Enterochromaffin cells in intestine</li> </ul>           | <ul style="list-style-type: none"> <li>• Accelerates gastric emptying</li> <li>• Increases movements of small intestine</li> <li>• Increases peristalsis in colon</li> </ul>                                                                          |
| <b>Substance P</b>                 | <ul style="list-style-type: none"> <li>• Brain</li> <li>• Small intestine</li> </ul>                                                           | <ul style="list-style-type: none"> <li>• Increases movements of small intestine</li> </ul>                                                                                                                                                            |
| <b>Ghrelin</b>                     | <ul style="list-style-type: none"> <li>• Stomach</li> <li>• Hypothalamus</li> <li>• Pituitary</li> <li>• Kidney</li> <li>• Placenta</li> </ul> | <ul style="list-style-type: none"> <li>• Promotes growth hormone (GH) release</li> <li>• Induces appetite and food intake</li> <li>• Stimulates gastric emptying</li> </ul>                                                                           |